

Deep Learning



Over-Squashing vs. Adversarial Robustness in GNNs An Add vs. Remove Edge Investigation



Presented by:

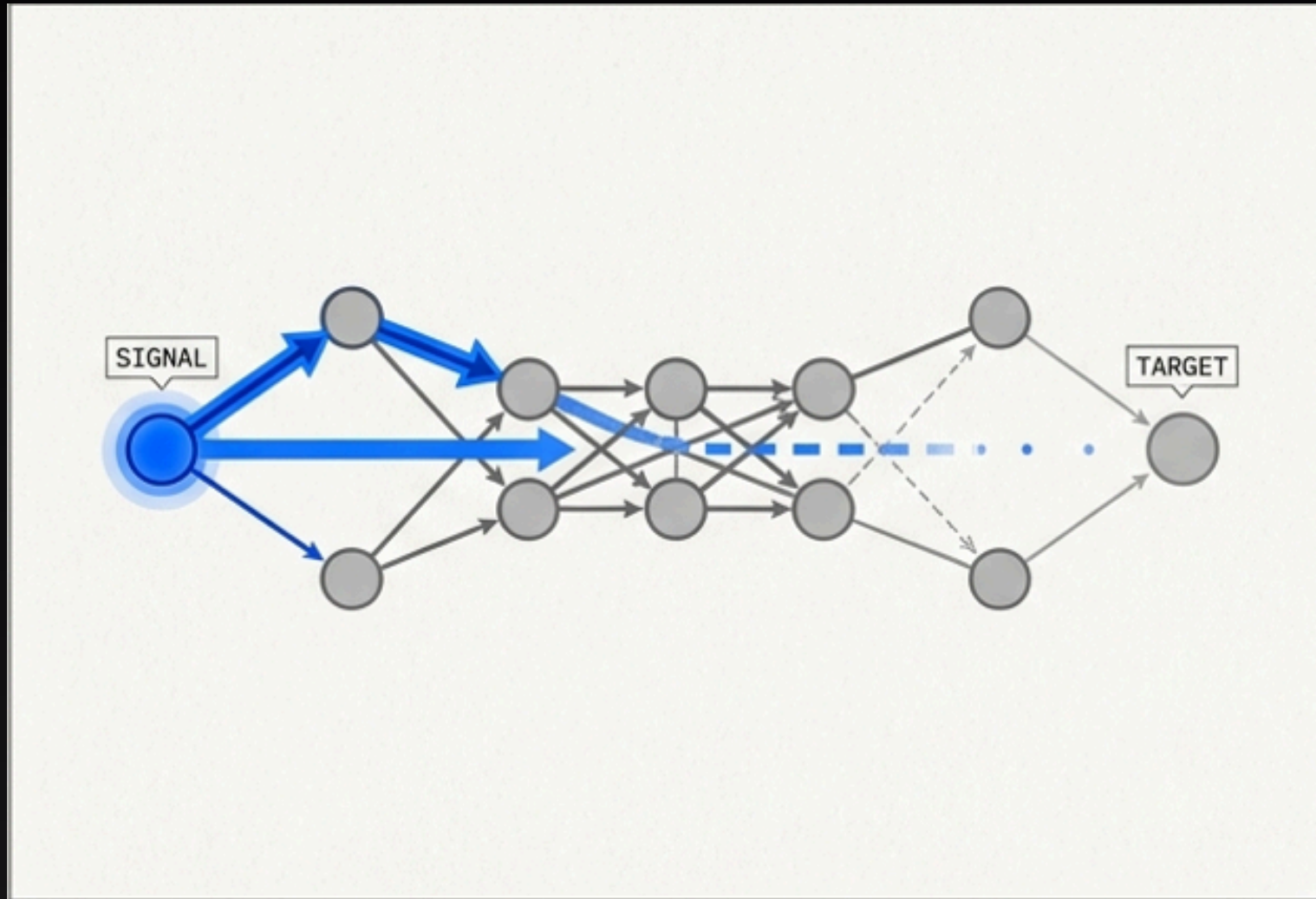
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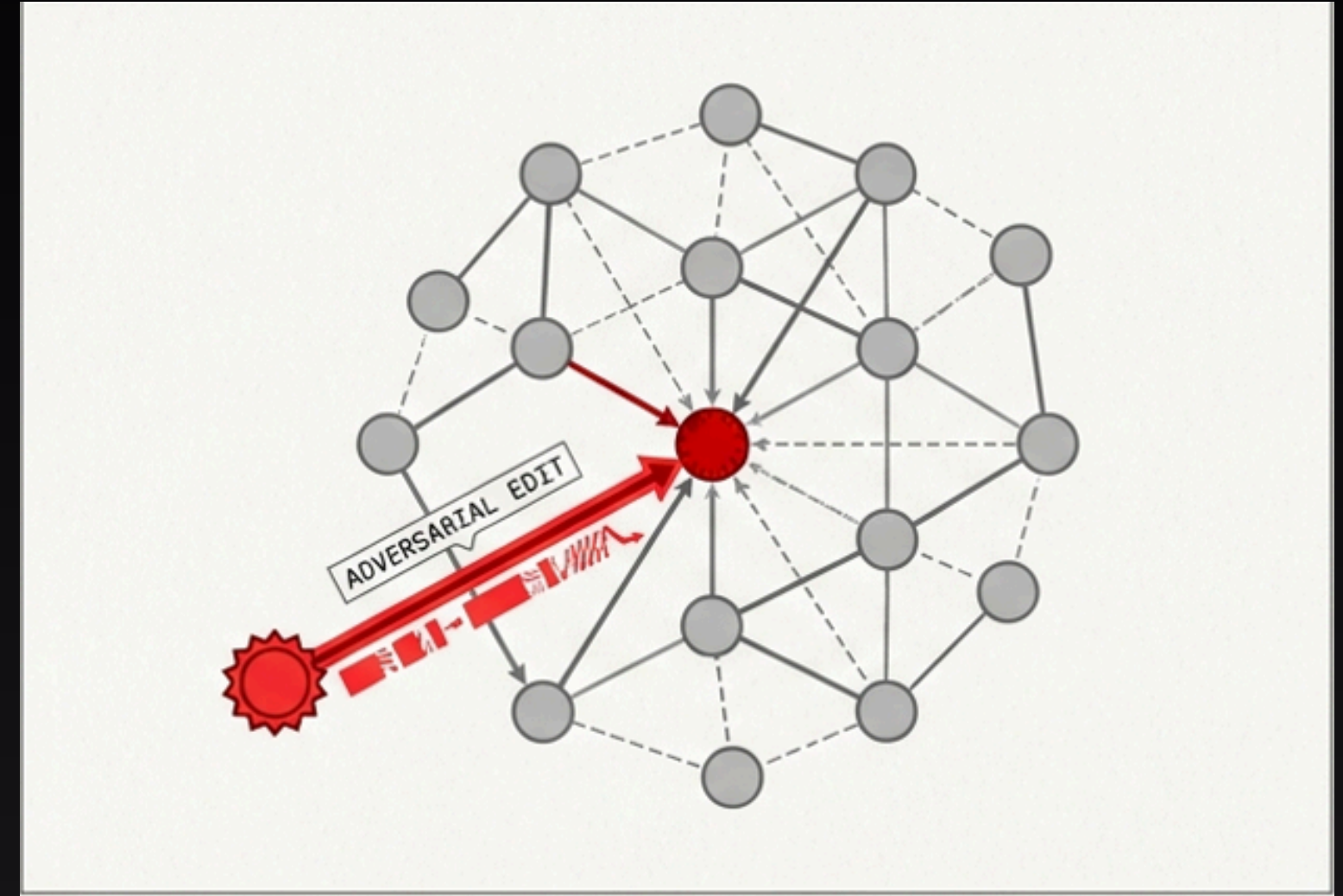
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Two Structural Failure Modes



Over-Squashing : Long-range information is compressed through bottlenecks and lost



Adversarial Fragility : A few maliciously edited edges flip predictions

Both pathologies fundamentally concern the same mechanism
“The sensitivity of a node’s output to structural changes elsewhere in the graph”

Over-squashing rewiring methods manipulate the graph structure.

- Over-squashing rewiring: FoSR **adds** edges; spectral pruning **removes edges** (both raise the spectral gap).
- Adversarial attack: adds dissimilar edges. Defense (GCN-Jaccard): **removes** them

Hypothesis:

- adding to fix over-squashing may hurt robustness (more edges to exploit);
- removing for over-squashing may transfer to robustness (removes like a defense).



Research Question

- RQ2 (main) Do over-squashing fixes (add-based FoSR, remove-based spectral pruning) improve adversarial robustness – vs. an undefended GCN and vs. a purpose-built defense – and at what cost to clean accuracy?
- RQ1 Does the attack worsen the over-squashing itself? (perturbation vs. long-range information loss)
- RQ3 (stretch) Can a single measure (Jacobian sensitivity) capture both over-squashing and adversarial fragility?

Related work

- The building blocks already exist – over-squashing rewiring (FoSR, spectral pruning), the attack (Metattack), the defense (GCN-Jaccard). We introduce no new method.
- The link is understudied – to our knowledge, rewiring/over-squashing surveys do not discuss adversarial robustness, and the robustness literature does not engage over-squashing.

⇒ Our contribution: a controlled add-vs-remove study connecting the two lines of work, reporting only real, reproducible numbers.

Methods

Methods on the add/remove axis

Method	Targets	Edge op	Role
GCN (vanilla)	—	none	baseline
FoSR	over-squashing	add	the “add” arm
ProxyDelete	over-squashing	remove	the “remove” arm
GCN-Jaccard	defense	remove	defense (similarity)
GCN-SVD	defense	low-rank	defense (low-rank)
DropEdge	regularization	random	sanity baseline

FoSR vs ProxyDelete = direction

ProxyDelete vs Jaccard = Criterion

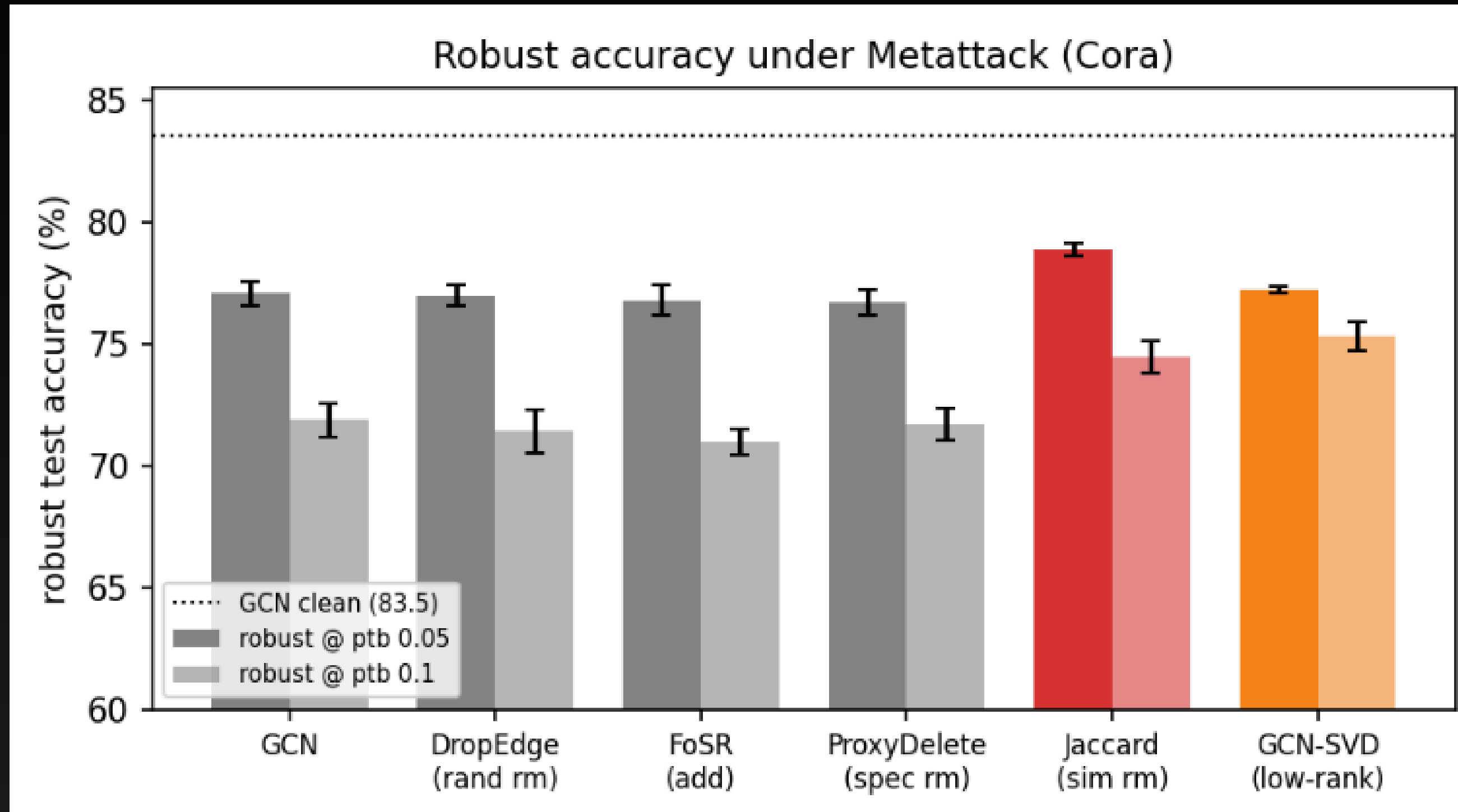
Setup

- Cora & Citeseer node classification, DeepRobust nettack split, CPU.
- Attack: Metattack (poisoning), ptb {0.05,0.10}; cached once, reused.
- Pipeline: clean → Metattack → method → train GCN → eval.
- Metric: clean / robust accuracy, robustness gap = clean – robust.

Datasets (standard nettack split)

Dataset	Nodes	Classes	Features
Cora	2,485	7	1,433
Citeseer	2,110	6	3,703

Results: add vs remove under attack (Cora)



Why does Jaccard win?

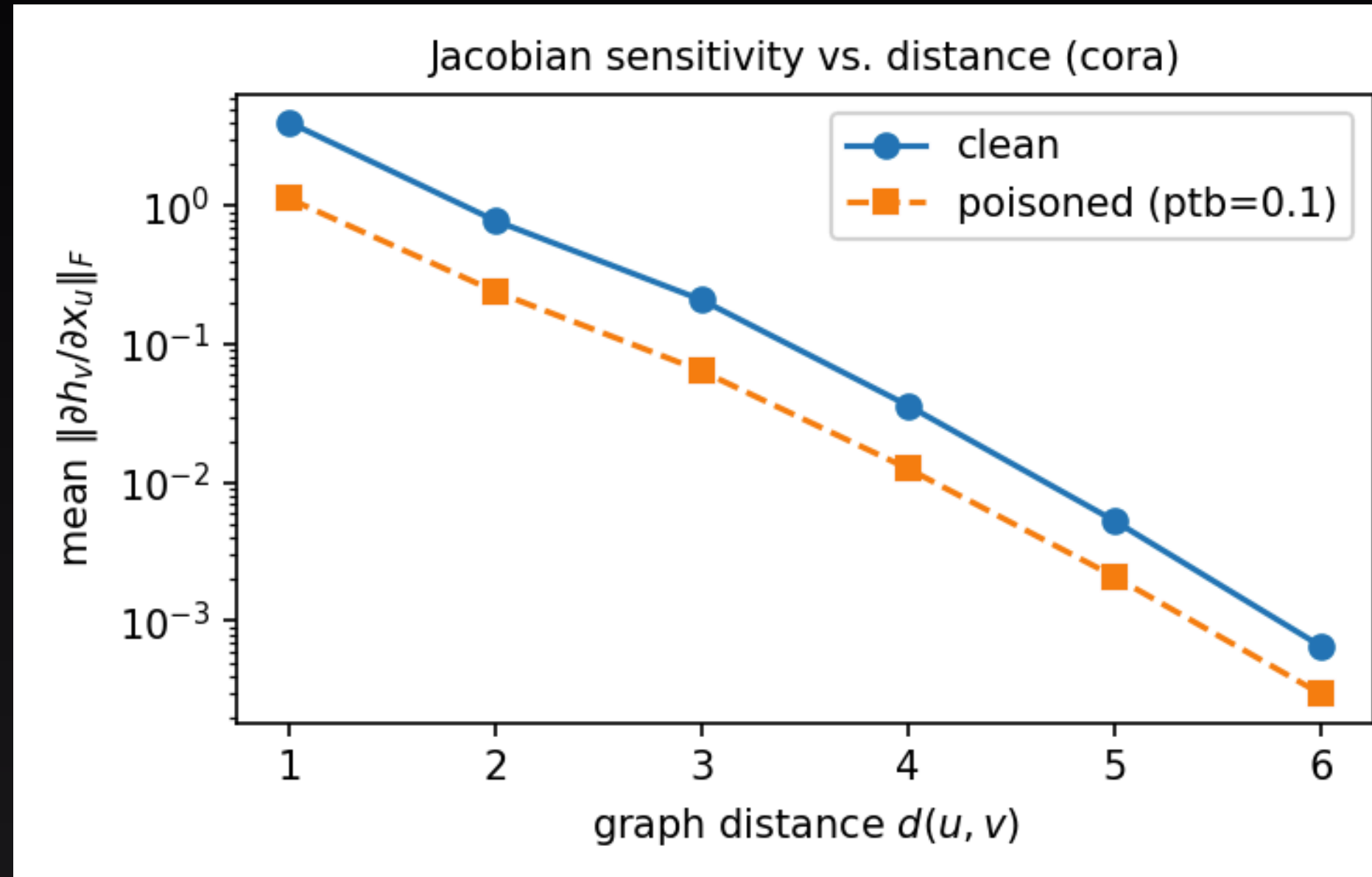
- Metattack almost purely adds edges (503 added, 3 removed at ptb 0.10). How many does each method delete?

Why Jaccard wins: attacker edges each method deletes (ptb 0.10)

Method	# removed	attack edges rm.	recall	chance
GCN-Jaccard	797	250	0.50	0.14
ProxyDelete (spectral)	643	93	0.19	0.12
DropEdge (random)	643	59	0.12	0.12
FoSR (add)	0	0	0.00	—

Chance recall ≈ 0.12 . Jaccard deletes half the attack edges ($\sim 4\times$ chance) $\Rightarrow$ it undoes the attack.
Spectral/random $\approx$ chance; add removes none.

Over-squashing probe (RQ1)



Sensitivity decays $\sim 5\times/\text{hop} \Rightarrow$ over-squashing is real. Poisoning shifts the level but not the decay rate $\Rightarrow$ over-squashing & fragility are decoupled (matches RQ2).

Adaptive attacker: evadable but self-defeating

A Jaccard-aware attacker adds edges between feature-similar nodes (the filter won't remove them).

- Evasion works at the filter: Jaccard removes 0% of these edges.
- But the attack turns weak: GCN drops only 0.7/2.1 pts (vs 6.5/11.7 for Metattack).

⇒ Edges that fool the model (dissimilar) are the ones Jaccard removes; edges that evade Jaccard (similar) barely fool it – **potency & detectability are coupled.**

Heuristic (not gradient-optimal) adaptive attack; optimal adaptive = future work

Findings

Ablation summary: which “knob” decides adversarial robustness?

Factor we varied	Held fixed	Result	What we learned
Add vs. remove (FoSR vs. ProxyDelete)	over-squashing goal	both fail to defend	direction is not what matters
Edit criterion (spectral / similarity / low-rank)	that we edit edges at all	only attack-aware editing defends (Jaccard, SVD)	the criterion is the key
Edge budget (253 vs. 643 edges)	the method	spectral removal still fails	it is not a budget artifact
Dataset (Cora vs. Citeseer)	the method	same ranking of methods	the finding generalizes
Attacker (non-adaptive vs. adaptive)	the defense	evades the filter but turns weak	potency & detectability are coupled
Over-squashing measure (clean vs. poisoned graph)	the model	decay rate unchanged	over-squashing & fragility are decoupled

Conclusion

- Over-squashing mitigations (FoSR add, ProxyDelete spectral-remove) do not improve adversarial robustness on Cora / Metattack.
- Only attack-aware defenses defend – Jaccard (similarity) & SVD (low-rank); robustness comes from targeting the attacker's edges, not from optimizing the spectral gap.
- Honest negative result: the over-squashing → robustness bridge is not supported here.
- Robust to budget (spectral removal still fails at 643 edges) & replicated on Cora + Citeseer. A defense-aware attacker evades Jaccard only by turning weak.
- Next: gradient-optimal adaptive attack, RQ3 (Jacobian → fragility).